Answers and talk notes

Multiply Decimals

What your student is learning

Tonight we multiply decimals to find a total cost, and we check whether the answer is reasonable before we trust it.

Answers

Ask your student to explain before you look.

1. **B.** About $12. Round 4.1 to 4 and $2.95 to $3: 4 × $3 = $12. About $12.
2. **C.** 3. 1.6 has 1 place and 0.25 has 2 places, so 1 + 2 = 3.
3. **A.** 10.5. 25 × 42 = 1,050 with 2 decimal places is 10.50, so 2.5 × 4.2 = 10.5.
4. **B.** $6.30. 15 × 420 = 6,300 with 1 + 2 = 3 decimal places is 6.300, which is $6.30.
5. **C.** $7.80. Estimate: about 2.4 × $3 = $7 to $9. The exact product is 240 × 325 = 78,000 with 3 decimal places, $7.80.

Talk about it

How did your estimate help you place the decimal point?

Home mission

Next time you shop, pick something sold by weight. Ask, "About how much for 2.5 kilograms?" Then ask, "Does the number on the scale make sense?"

When they are stuck, ask

- About how much should this be? Estimate before you multiply.
- How many decimal places in the first factor? And in the second?
- Does your answer match your estimate, or is it ten times off?

Try not to say

- Count the decimal places in the bigger number only.
- Line up the decimal points like you do when adding.
- Skip the estimate, it just takes extra time.